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The Rivermead Post-Concussion Questionnaire score is associated with disability and self-reported recovery six months after mild traumatic brain injury in older adults

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ABSTRACT

Background/Objectives: Post-concussion syndrome refers to the adverse group of symptoms following a mild traumatic brain injury (mTBI). The Rivermead post-concussion syndrome questionnaire (RPQ) is a common clinical tool for assessing baseline post-concussion syndrome symptomatology; however, it is unknown if scores on this questionnaire are associated with future disability. Therefore, the goal of this study was to determine the association between baseline RPQ scores and future disability in older adults with mTBI.

Methods and Findings: This study used a prospective cohort design, using the RPQ to measure baseline post-concussion syndrome symptomatology. Disability at 6 months was measured using the Glasgow Outcome Scale-Extended (GOSE; disability), short-form 12 (SF-12; physical and mental quality of life), and self-reported recovery. Linear and logistic models adjusted for confounding factors were estimated using 200 bootstrapped samples. Individuals with higher levels of baseline symptomatology were more likely to have poor GOSE scores (RR = 2.13, 95% CI [1.51, 2.31]) and self-reported recovery (RR = 2.64, 95% CI [1.31, 8.98]) 6 months later.

Conclusions: High levels of baseline symptomatology may be associated with overall disability and individual perceptions of recovery 6 months post-mTBI. While the RPQ is valid in assessing a patient's post-concussive symptoms following mTBI, it may not predict long-term physical or mental health in older adults.

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Introduction

Mild traumatic brain injury (mTBI) is a prominent health concern, with enormous costs and burden to society. Approximately 70-90% of traumatic brain injuries are mild in nature (1-3). Current estimates of the incidence of mTBI suggest that the condition affects 300-600 per 100,000 individuals annually (4,5). However, it is likely that these figures grossly underestimate the true prevalence rate, as many individuals do not seek professional medical treatment following a concussion (2-6).

While mTBI is a health concern across all age cohorts, a particular interest exists for mTBI in older adults (7). As the population of developing nations continues to age, focusing health studies on older persons should be a research priority (7). After youth and young adults, elderly adults suffer the highest rates of mTBI through mechanisms such as falls (4,8). While age has been cited in the literature as a risk factor for mTBI, few studies have addressed this population specifically, and few have examined prognosis in this age cohort following mTBI (7,8).

Post-concussion syndrome is the medical term given to the adverse group of symptoms that can arise following mTBI that

consists of an array of physical, emotional, and cognitive impairments (2,6). Generally, these symptoms tend to persist for approximately 7 to 10 days following mTBI; however, some individuals may experience this post-concussive symptomatology for three to 12 months following injury (2,9). Studies have indicated that 50% of individuals' experience post-concussion syndrome symptoms at 1 month post-injury, and as many as 25% may still have persisting symptoms at 1 year and beyond (1,2,9,10). There is substantial literature in regard to the burden of post-concussion symptomatology at baseline and recovery. In general, individuals who report more symptoms at baseline in relation to their concussion are more likely to report post-concussion symptomatology later (8,11,12). Older adults may feel a greater burden of post-concussion syndrome symptoms than their younger counterparts due to age-related complications (11).

As the burden of the post-concussion syndrome varies between individuals, it becomes of interest to establish factors that may be used to identify individuals who experience longer-lasting adverse symptoms (12). By recognizing these higher-risk individuals, clinicians are better able to tailor care plans and improve quality of life (QOL). Unfortunately, few studies have investigated the long-term consequences of

mTBI, and even fewer have examined mTBI recovery and prognosis in elderly adults, despite the potential clinical benefits of conducting such studies (7,9,13,14).

The Rivermead Post-Concussion Questionnaire (RPQ) is a valid test commonly used in the clinical setting to assess cognitive impairment in individuals following mTBI (15). The RPQ is a well-established assessment tool of post-concussion syndrome symptoms that may be self-administered, or administered by a physician in the hours or days following initial mTBI (15). In principle, high scores on the RPQ suggest that the individual suffers from a high burden of post-concussion syndrome symptoms at the time of test administration. From a logical standpoint, a baseline assessment of these symptoms could potentially provide clinicians with a tool to identify patients who may recover more slowly. Other studies concerned with post-concussion symptomatology have looked at the burden of symptoms at baseline, and have frequently found that individuals reporting more symptoms at baseline tend to have more symptoms at a later time (8,11,12). It is currently unknown if this same association would be shown with the RPQ, as it has had limited application in prognosis for mTBI.

Determining whether specific RPQ scores are associated with worsening future outcomes is potentially of high clinical significance (14). In a previous pilot study we conducted, we determined that poor RPQ scores, indicating high levels of symptomatology, are correlated with a variety of poor recovery markers, yet it is not clear if RPQ scores are associated with recovery when controlling for other potentially confounding factors (15). Therefore, the primary objective of this study is to determine the association between initial RPQ scores and future disability in older adults with mTBI, while controlling for confounding factors. Our goal is not to predict future disability, but rather to explore the associations between baseline RPQ score and four distinct recovery outcomes. These recovery outcomes include the Glasgow Outcome Scale Extended (GOSE), the Short Form-12 physical and mental components (SF-12 PCS and MCS) and self-reported recovery. We hypothesized that high scores on the RPQ at baseline will be associated with poor recovery outcomes 6 months later.

Methods

Study design

This study used a prospective cohort design, assessing post-concussion syndrome symptomatology at baseline and recovery outcomes six months later. The study design was previously described in a publication describing the association between various prognostic factors and RPQ as an outcome (15). Briefly, the study was carried out in three emergency departments (ED) located in Ontario, Canada. The project received Research Ethics Board (REB) approval from Lakehead University (Project # 035 12–13), (Project #99 2012116), and the Thunder Bay Regional Health Sciences Centre (Project #EMED-181-12).

Definition of mTBI

For the purposes of this study, we used the WHO definition of mTBI that defines mTBI as 1) a mechanical injury to the

head resulting from external physical forces resulting in 2) one or more of the following: confusion or disorientation, loss of consciousness for 30 min or less, post-traumatic amnesia (PTA) for less than 24 h, and one or more of a variety of temporary neurological abnormalities (i.e. seizure, dizziness, memory problems, etc.). Finally, 3) an mTBI is considered mild as long as the individual presents with a Glasgow Coma Scale (GCS) score of 13–15 at 30 min or later post-injury (7). All mTBI diagnoses were made by ED physicians using the above definition.

Eligibility criteria

All individuals aged 65 or older who had visited a participating ED between 5 November 2012 and 13 November 2013 for mTBI were potentially eligible for inclusion in the study. To be eligible, the individual had to be an Ontario resident and seek medical attention within 72 h of sustaining an mTBI (i.e. stand-alone injury, part of multi-system trauma, etc.). In cases where the inclusion criteria were met, individuals were contacted by phone within 10 days of their hospital visit and invited to participate. Individuals who were unable to provide information due to illness or lack of comprehension at any phase of the study (including follow-up) were excluded. Individuals with a CT scan that revealed large intracranial bleeds (i.e. subdural hematoma, epidural hematoma, subarachnoid hemorrhage) as diagnosed by radiologists were excluded. Individuals with less severe CT head findings were included. These findings include edema, smear subdural hematoma and/or contusion. CT scans were not a requirement nor disqualifier for patients of this study, so long as major bleeds were not identified. Further, the manifestations of mTBI could not be due to drugs, alcohols, medications, caused by other injuries or treatments related to other injuries or by other problems (such as psychological trauma).

Data collection

We provided an information and consent form to potential participants in person at the time of ED visit or by mail the next business day. Participants completed a baseline computer-assisted telephone interview (CATI) survey within 10 days of ED visit. If after five phone call attempts we were unable to reach the potential participant, the individual was considered to be a non-respondent. At 6 months post-injury, participants were contacted to complete another CATI survey to assess their recovery. If after five attempts, we were still unable to contact a participant; he or she was considered lost to follow-up.

Clinical data were extracted from the ED chart for participants who provided verbal consent at the time of the baseline interview. Where recorded, we extracted information on date of injury, date of visit to ED (if different from date of injury), location of injury (for example, home), activity during injury, mechanism of injury, GCS, PTA duration, LOC duration, CT characteristics, early symptoms (such as dizziness, nausea/vomiting, headache), medication use, and physician rating of global recovery. Other concurrent injuries were also recorded.

Post-concussion syndrome measurement

We used the RPQ to measure baseline post-concussion syndrome symptomatology. The RPQ is a valid measure that assesses patients on 16 typical symptoms (16). The test was administered by the research team over the phone, and the patient was asked to base their ranking on the severity of the symptom as compared to its severity prior to the head trauma event. Several other studies have indicated strong reliability and validity of this questionnaire, but with no specific focus on an elderly population (16–19). The RPQ consists of 16 symptoms of post-concussion syndrome. For each of the 16 symptoms, the individual responded on a Likert-scale with scores ranging from 0 to 4 on how much each symptom had impacted them since their injury (0 = not experienced at all, 1 = a minimal problem, 2 = a mild problem, 3 = a moderate problem, and 4 = a severe problem) (16–18). While various means of calculating an individual's RPQ score exist, for the purposes of this study, the RPQ was determined as a score out of 16, representing the number of symptoms on which the patient was considered to be “non-recovered”. For each symptom, individuals scoring 0 or 1 (those who did not experience the symptom or experienced it minimally) were considered “recovered”, and individuals scoring 2, 3, or 4 (those who did experience the symptom more than a minimal amount) were considered “non-recovered” (15,20). Using this continuous measurement scheme, a higher score indicated more post-concussion syndrome symptoms.

Recovery measures

Recovery measures were determined at 6 months post-injury. These measures of disability included the Glasgow Outcome Scale Extended (GOSE), the Short Form-12 physical and mental components (SF-12 PCS and MCS) and self-reported recovery.

Glasgow Outcome Scale-Extended version (GOSE)

The GOSE was used as an assessment of overall disability. The GOSE consists of an 8-point, hierarchical scale that provides an assessment on an individual's independence and functioning following a head injury (21–23). Nineteen questions contained within eight broad categories of global disability were posed to the participants, and they ranked themselves as: severely disabled (unable to support themselves for 24 h in society), moderately disabled (significant restrictions in lifestyle or work capacity, or both), or having good recovery (have resumed their previous lifestyle) (21). An individual's overall score was determined by taking the lowest score (1–8) from all of the questions posed (22). A score of 8 indicates good recovery in functioning, while a score of 1 indicates patient death (22). For the purposes of this study, only individuals scoring 7 or 8 were considered to have “good recovery”. Individuals who scored less than 7 were considered to have “poor recovery”. The GOSE has shown adequate test-retest, inter-observer reliability, validity, and good responsiveness in an adult population (23–27).

SF-12

The SF-12 is a short-form, multipurpose health survey that assesses various generalized components of health-related

quality of life (28). The short-form version of this test contains 12 questions selected from a 36 question survey, aimed to reduce the burden on the patients completing the survey (28). The questions are combined, scored, and weighted to create two distinct recovery measures: the Physical Composite Score (PCS) and the Mental Composite Score (MCS) (28). These two scores, each on a scale of 0–100 (with 100 indicating the best possible recovery), provide interpretable measures of physical and mental well-being following injury (28,29). We reverse scored the PCS and MCS, such that 0 indicated the best possible recovery, to model poor recovery instead of a good recovery. The SF-12 has been tested for reliability and has been deemed to be valid in older adults (28,30).

Self-reported recovery

This measure is a self-perceived method of assessing recovery that asked the participants to rank, based on comparison with pre-injury status, how well they perceive their state of recovery. Self-reported recovery is considered to be a reliable, valid and responsive measure of self-perceived well-being (31). Of note, it has not been validated in an elderly adult mTBI population. Participants' perception of their own recovery was measured by asking a question: “How well do you feel you are recovering from your injuries?” Participants who responded “all better” or “quite a bit of improvement” were considered to be recovered. The other choices of self-reported recovery (“some improvement” and “no improvement”) were designated as “not recovered”, and considered the outcome of interest.

Confounding variables

Potential confounding variables were identified as factors significantly associated with each recovery outcome of interest, as acknowledged in our previous study (15).

Health 1 year before injury

This question was posed to participants to rate their health at the time of injury to their general state of health 1 year prior to the injury. Participants were able to rate their change in health on a Likert scale ranging from 1 = much better to 5 = much worse.

Loss of consciousness

This question was included as a self-report question that asked participants if they had lost consciousness after sustaining their mTBI.

Center for epidemiologic studies depression scale (CES-D)

The CES-D is a twenty question self-report survey first developed in 1977 by Laurie Radloff (32). The CES-D is a validated and reliable self-report tool and is widely used to collect information regarding an individual's self-reported burden of depressive symptoms (32,33). Scores range from 0 to 60, with higher scores indicating more depressive symptoms. The CES-D also displays high test-retest validity and good internal consistency (31–37).

Beyond the variables mentioned above, other potential confounding variables that were assessed but not included were fatigue, visiting another health-care provider and consulting another health-care provider.

Data analysis

All data were analyzed using Stata statistical software version 14.0 (38). Statistical significance was set at $p < .05$.

The counts and percentages were used for describing categorical data. The mean and standard deviation (SD) were used to summarize the quantitative variables. Individuals who were missing outcome measures at the 6-month follow-up were removed from the analysis. The bivariate correlations were used to look for the basic nature, strength and direction of relationships between the dependent (SF-12 PCS and MCS, GOSE and self-reported recovery at 6 months) and independent variable (baseline RPQ). Simple linear regressions were conducted to determine the association between SF-12 PCS and MCS outcomes 6 months post-injury per unit increase in the baseline RPQ measure. Logistic regression was used to determine the risk of overall disability (as measured by GOSE) and poor self-reported recovery associated with baseline RPQ. All final regression models were conducted using bootstrapping with the replication number set to 200. Potential confounding variables were identified from a previous study as those highly associated with the outcome of interest (15). Each potential confounder was added to the crude model and those whose addition resulted in a 10% change in effect size were included in a final model. Final effect estimates were calculated based on models with important confounding variables included. Associations are reported as regression coefficients or relative risks (RR) and 95% confidence intervals, as appropriate. Relative risks were computed indirectly from multivariable logistic regression models using the procedure outlined by Localio, Margolis & Berlin (2007) (39). For multiple regression models with SF-12 PCS and MCS as continuous outcomes, normality testing for these dependent variables was performed using Shapiro Wilk test. We also computed the variance inflation factor (VIF) as an additional measure to look for multicollinearity among the independent variable (RPQ) and confounders and a VIF below 10 was considered acceptable. For logistic regression models with GOSE and self-reported recovery, the omnibus test was used to look for the significance of the model coefficient. The Hosmer and Lemeshow test was used to evaluate the goodness of fit for the models.

Results

Participants and demographics

Of 97 eligible participants, 54 agreed to participate (55.7% participation rate). Of these 54, five were lost to follow-up. An additional three participants were removed due to the occurrence of another head trauma over the 6-month follow-up period. The final sample available for analysis included 46 persons. Available baseline characteristics of the participants, non-participants, and those lost to follow-up are displayed in Table 1. Reasons for the five lost to follow-up included one who did not want to respond further, two had poor health and moved to a long-term care facility, one's spouse had recently died, and for the last, a reason was not provided. The mean baseline RPQ score for the followed sample ($n = 46$) was 3.20 while the mean baseline RPQ score for the five lost to follow-up was 14.4,

Table 1. Baseline characteristics of older adults visiting the emergency department for MTBI (full participants, those lost to follow-up, and non-participants).

Characteristic	Full Participants (N = 46) (%)	Participants lost to follow-up (N = 5) (%)	Eligible non- participants (N = 43) (%)
Age group			
65 to < 75 years	45.7	20.0	27.9
75 to < 85 years	43.5	20.0	23.3
≥ 85 years	10.9	60.0	23.3
Missing	0.0	0.0	25.6
Sex			
Male	41.3	60.0	34.9
Female	58.7	40.0	65.1
Education			
High school or less	47.8	40.0	
Post-secondary, Technical school, or some university	17.4	20.0	
University graduate	32.6	0.0	
Don't know/Declined	2.2	40.0	
Income (\$CDN)			
\$0 to \$40,000	37.0	0.0	
\$40,001 to \$60,000	19.6	20.0	
\$Above \$60,000	26.1	0.0	
Don't know/Declined	17.4	80.0	
Employment status			
Retired	82.6	100.0	
Full time	4.3	0.0	
Part time	6.5	0.0	
Self employed	4.3	0.0	
Homemaker	2.2	0.0	
Marital status			
Single	8.7	0.0	
Married/Common Law	65.2	40.0	
Widowed	19.6	20.0	
Separated/Divorced	6.5	20.0	
Don't know/Declined	0.0	20.0	
Ethnicity			
White	97.8	80.0	
Black	2.2	0.0	
Don't know/Declined	0.0	20.0	
BMI			
Healthy weight	39.1	40.0	
Underweight	4.3	0.0	
Overweight	32.6	20.0	
Obese	19.6	0.0	
Missing	4.3	40.0	
Previous Head Injury			
Yes	30.0		
No	70.0		

indicating that those lost to follow-up had more severe symptoms at baseline compared to those who were followed up at 6 months. The majority of the sample population was aged 65–85 years, were female, did not screen positive for subclinical depression, and had, on average, three symptoms of post-concussion syndrome as screened by the RPQ. The baseline RPQ assessment was completed within 10 days of ER visit. The range of days between ER visit and contact by the team ranged from 1 to 10, with a mean of 4 days. Baseline and follow-up values of the RPQ, GOSE, SF-12 PCS and MCS, as well as self-reported recovery are reported in Table 2. At follow-up, the mean reported that the number of unfavorable symptoms on the RPQ was reduced to 1.17 (Table 2).

Crude associations between the RPQ at baseline and each outcome of interest are reported in Table 3. The crude associations found between baseline RPQ and GOSE, and SF-12 mental health component suggested that overall disability and mental health outcomes 6 months later are associated with more RPQ symptoms at baseline. The crude association

Table 2. Summary Statistics for Variables of Interest (Participants included in analysis); $n = 46$.

Variables	Baseline				6 month follow-up			
	Mean or proportion	95% CI	Median	Range	Mean or proportion	95% CI	Median	Range
RPQ*	3.20	2.21-4.18	2.0	0-13	1.17	0.47-1.87	0	0-10
GOSE**	56.5%	41.6-71.4			15.2%	4.43-26.0		
SF-12 PCS***	51.6	43.09-60.10	56.3	0-100	22.74	16.15-29.32	16.7	0-83.3
SF-12 MCS****	27.6	21.34-33.79	25.4	0-86.7	14.84	10.38-19.30	10.0	0-67.5
Self-reported recovery*****	78.3%	65.9-90.6			23.9%	11.1-36.7		
Loss of consciousness	24.4%	10.7-38.1			N/A			
Health one year before injury $\pm$	3.09	2.90-3.27	3.0	1-5	N/A			
CESD score $\pm$	7.57	5.32-9.81	5.0	0-36	6.41	3.82-9.01	3.0	0-39

*RPQ score, ranging from 0 to 16, representing the number of symptoms on the RPQ with an unfavorable outcome.

**Proportion non-recovered

***SF-12 PCS refers to an individual's SF-12 Physical Component Score, ranging from 1 to 100. A score of 0 indicates perfect recovery, while a score of 100 indicates poor recovery.

****SF-12 MCS refers to an individual's SF-12 Mental Component Score, ranging from 1 to 100. A score of 0 indicates perfect recovery, while a score of 100 indicates poor recovery.

*****Proportion non-recovered

$\pm$ Health 1 year before injury, with scores ranging from 1 (much better than 1 year ago) to 5 (much worse than 1 year ago)

$\pm$ CES-D subclinical depression scale, with scores ranging from 0 to 60. A score of 0 indicates no depressive symptomatology and a score of 60 indicates maximum depressive symptomatology.

Table 3. Unadjusted and adjusted associations between baseline RPQ scores and various recovery outcomes assessed at 6-month follow up.

Recovery outcome (modeling poor recovery)	Control Variables (assessed at baseline)							
	Unadjusted		Health one yr before injury		Health one yr before injury + CESD		Health one yr before injury + CESD + LOC	
	N	Estimate (95% CI)	N	Estimate (95% CI)	N	Estimate (95% CI)	N	Estimate (95% CI)
SF 12 PCS	46	B = -0.38 (-1.65-2.41)	46	B = -0.06 (-1.89-2.02)	46	B = 0.09 (-2.21-2.38)	41	B = -0.34 (-2.19-2.86)
SF 12 MCS	46	B = 1.42 (0.11-2.73)	46	B = 1.21 (-0.47-2.88)	46	B = 0.36 (-1.37-2.10)		N/A
	Unadjusted		LOC		LOC + CESD		LOC + CESD + Health one yr before injury	
	N	RR (95% CI)	N	RR (95% CI)	N	RR (95% CI)	N	RR (95% CI)
Self-reported recovery	46	RR = 1.75 (-0.07-3.56)	41	RR = 2.62 (1.54-6.03)	41	RR = 2.64 (1.38-7.06)		N/A
GOSE	46	RR = 1.68 (1.25-2.31)	41	RR = 2.13 (1.51-6.07)		N/A		N/A

RPQ Rivermead Post-Concussion Questionnaire; SF12 PCS 12-item Short Form Health Survey-Physical component summary; SF 12 MCS 12-item Short Form Health Survey mental component summary; GOSE Glasgow Outcome Scale-Extended Version; B unstandardized Beta coefficient; CI confidence interval; N/A Not applicable

between baseline RPQ and SF-12 physical health component and self-reported recovery did not show any significant associations.

Adjustment for confounding

Three potentially significant confounders were identified from our previous study (15). Confounders that caused an effect size change of more than 10% were included in the final model (Table 3). The maximum number of confounders found for the SF-12 physical health model was three. These confounders included health 1 year prior to the injury, sub-clinical depression (as determined through the CESD score), and LOC. In the association between baseline RPQ and GOSE, the only confounder found to cause a significant change (>10%) in the effect estimate was loss of consciousness (LOC) at the time of injury. For the association between baseline RPQ and SF-12 mental health component, only health 1 year before the injury and baseline depression scores were deemed important confounders. Finally, the association between baseline RPQ and self-reported recovery included LOC and baseline depression as confounders.

For the adjusted linear regression models for SF-12 PCS and MCS, regression diagnostics suggested that there was no multicollinearity among the independent variable and confounders. Graphically, residuals were normally distributed. For the adjusted logistic regression models for GOSE and self-reported recovery, the omnibus test of the model coefficient suggests that independent variables and confounders significantly ($p < .05$) contributed to the model. Hosmer Lemeshow goodness of fit test was not significant ($p > .05$) for the regression models, suggesting good fit of our logistic regression models for GOSE and self-reported recovery.

When full, adjusted models were run, no significant associations were found between baseline RPQ and six-month SF-12 PCS and MCS scores (Table 3). In comparison, a significant association was found between baseline RPQ and six-month GOSE scores. The adjusted relative risk (RR = 2.13 (95% CI: 1.51, 6.07)) for the association between baseline RPQ and GOSE 6 months post mTBI suggested that those with more RPQ symptoms at baseline had a poorer overall disability (as measured by GOSE) than those who had fewer RPQ symptoms at baseline. The adjusted relative risk (RR = 2.64 (95% CI: 1.31, 8.98)) for the association between RPQ at baseline and self-reported recovery at 6 months suggested that those with more

unfavorable RPQ symptoms at baseline were more than twice as likely to feel as though they had not recovered at 6 months than those with favorable RPQ symptomatology at baseline. These models with significant findings suggest that individuals with higher RPQ symptomatology at baseline had a worse global disability and self-reported recovery at 6 months post-injury.

Discussion

To our knowledge, this is the first study that evaluates the association between baseline RPQ score and recovery 6 months after mTBI in older adults. In this study, we found that individuals with greater baseline symptomatology, according to the RPQ, were more likely to have increased global disability and poorer self-perceived recovery at 6 months post-injury. Our findings are similar to a recent study published in December 2017, where cognitive and psychological functioning was found to be impaired in individuals following mTBI (40). Further, our findings are similar to studies using different baseline measurements that have found associations between high baseline symptom reporting and increased post-concussion syndrome symptoms (11,12,40).

Interestingly, we found that there was no statistically significant association between RPQ score at baseline and physical and mental recovery as measured by the SF-12. Therefore, we conclude that the burden of post-concussion syndrome symptomatology is likely independent of the burdens an individual may face 6 months later in regards to physical and mental health. We also note that it is possible our sample was an exceptionally healthy group of individuals. However, we do not think that this is the case as the prevalence of subclinical depression in our sample was about 11%, which is keeping with the national estimate (41).

The three confounding variables that we considered in our different models included CESD (subclinical depression assessment), loss of consciousness, and health 1 year prior to the injury. Health 1 year prior to injury makes logical sense, as individuals who have had declining health in the year previous will likely recover later, and may have been more prone to sustaining mTBI in the first place. The confounding effects of depression, as measured through the CESD are of topical interest. There has been growing research on the association between depression and mTBI, and questions surrounding the potential synergistic effects these conditions may have on each other (42). CESD was a significant confounding variable in all of our models except for overall disability. Interestingly, it has been suggested that fear avoidance may be in part able to explain why individuals with post-concussion symptomatology may suffer from depression (42). This hypothesis also lends support to the notion that these individuals may be physically healthy in the months following their injury, but may see a decline in their overall global ability (if they are avoiding outings due to fear). Further research into the association between mTBI, depression, and fear avoidance is warranted.

Due to a high rate of falls, older adults represent an important population to target with tailored care plans designed at reducing the effects of post-concussion syndrome symptomatology following mTBI. Based on our findings, the initial burden of post-concussive symptomatology such as nausea, headache, and

dizziness had no association with physical recovery in elderly adults 6 months post-mTBI. Future studies could explore other tools that may help to assess the prognosis of elderly adults with mTBI in this respect. Our findings do suggest that a higher number of post-concussion syndrome symptoms following mTBI may indicate future disability and personal feelings regarding recovery. The broad disability domain covered by the GOSE scale may be of particular relevance to family practitioners, as they may anticipate that elderly adults with a high burden of post-concussion syndrome symptoms may likely require extra help in regards to mobility and overall functioning 6 months post-injury.

Limitations

A limitation of this study is the small sample size. However, despite the small sample size, we were still able to detect statistically significant results. Further, we were able to adjust for the effects of multiple confounders with statistical significance. These results are based on 200 replicate bootstrapped models. We are therefore quite confident that the associations reported in our study are stable, reliable and not due to chance.

As a result of a 55% participation rate, it is possible our study may contain selection bias and lack generalizability to the entire older adult population (specifically those greater than 85 years of age). It is quite possible that the individuals choosing to participate in the study may have had a systematic reason in choosing to do so. While we have basic demographic information for participants and non-participants, potential factors to explain participation are missing for the non-participants. Some of the confounding measurement tools we used have not been specifically validated in elderly populations. Further, there are likely to be other confounding factors given the nature of the data collection. For example, we do not have details regarding participants' health, hospital admissions and co-morbidities between the baseline and six-month follow-up. Finally, as this study was concerned exclusively with older adults, it should be noted that generalizations to the broader population should not be made.

Conclusion

While the RPQ remains a valid tool in assessing patients' post-concussive symptoms following mTBI, it may not be appropriate to predict long-term physical or mental health from its baseline RPQ scores in older adults. High levels of post-concussion symptomatology at baseline may be associated with worse psychosocial and self-perceived well-being as well as increased global disability. This information may help inform clinicians about the six-month prognosis of their older patients with mTBI.

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